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Technical Manual

Part I, Technical Info
Volume 1, Technical Description

INTELLIGENT VEHICLE INTERCOM SYSTEM - iVIS

(UCZN10331)

DS Cat. No. _____

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RECORD OF CHANGES

Change Notice No.	Authority	Incorporated by “Name & Rank” in Block Letter	Initial with Date
(1)	(2)	(3)	(4)

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ASSOCIATED HANDBOOKS

Technical Literature for **Digital Communication Harness iVIS UCZN10331**

comprises of the following:

- USER HANDBOOK
- TECHNICAL MANUAL

PART I TECHNICAL INFORMATION

Volume 1 **Technical Description**

Volume 2 Drawings

PART II MAINTENANCE

PART III OVERHAUL AND RECONDITIONING INSTRUCTIONS

PART IV MANUFACTURER'S PARTS LIST

Volume 1 Parts List

Volume 2 Illustrations

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SAFETY WARNING

The voltages employed in this Equipment are sufficiently high to
endanger human life.

**POWER
MUST BE SWITCHED OFF**
before servicing the equipment
&
GREAT CARE
taken when making internal adjustments etc.

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

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FIRST AID IN CASE OF ELECTRIC SHOCK

 Fig. 1	1.	SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BAREHANDS until he is clear of the conductor, but DO NOT WASTETIME .
 Fig. 2	2.	(a) Place the victim in supine position. (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it with water or saline mucus or blood, a lot of which might have accumulated in the back of the throat (Figure 1).
 Fig. 3	3.	If the jaw is rigid, try to force the mouth open by the pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages. (Figure 2-3).
 Fig. 4	4.	(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with other. (Figure 4). (b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG). (Figure 4)
	5.	When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical center.
HAVE SOMEONE ELSE AND SEND FOR A DOCTOR KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.		

CAUTION / NOTES

**Power from external power sources is
to be disconnected before servicing the equipment.**

Great care

**While connecting power from power source great
care shall be exercised for polarity of the power cables**

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE

(A) General

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER.**

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

- (i) Smash - Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers, crowbars and heavy tools.
- (ii) Cut - Cut the cords, headsets and wiring. Use axes, hands axes or hatchets.
- (iii) Burn - Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary.
- (iv) Bend - Bend the panels, cabinet and chassis.
- (v) Explosives - If explosives are necessary, use firearms, grenades or TNT.
- (vi) Disposal - Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams.

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Table of Contents

Chapter 1 – Brief Technical Specifications	17
1.1. About iVIS.....	17
1.2. Core Functionalities and Features of iVIS	17
1.3. Components of iVIS System	17
1.4. Design & Integration Characteristics	18
1.5. iVIS Central Switch Unit (CSU)	19
1.5.1. iVIS CSU – Key Functions and Capabilities:	20
1.5.2. iVIS CSU – Dimensions and Weight:	21
1.5.3. iVIS CSU – Ports and Peripherals	21
1.5.4. iVIS CSU – Interfaces:	23
1.6. iVIS User Unit (UU).....	24
1.6.1. iVIS UU – Key Functions and Capabilities.....	24
1.6.2. iVIS UU – Dimensions and Weight:	26
1.6.3. iVIS UU – Interfaces:	27
1.7. iVIS Headset Options	28
1.7.1. iVIS Headset – ANR Headset	28
1.7.2. iVIS Headset – PNR Headset	29
1.8. QWERTY Keyboard	30
1.9. Stick Radio	30
1.10. Stick Radio Dock Station	31
Chapter II – Brief Principles of Operation	32
Chapter III – Detailed Circuit Theory	34
3.1. About iVIS PCBA Cards:.....	34
3.2. iVIS Central Switch Unit (CSU) PCBA Cards:	34
3.2.1. iVIS CSU – iVIS EMI/EMC Power Filter Card.....	34
3.2.2. iVIS CSU – 10-Port Ethernet Switch with SPE Card	35
3.2.3. iVIS CSU – iVIS Dual Radio Card	35
3.2.4. iVIS CSU – Dual Radio Extension Card (Optional)	36
3.2.5. iVIS CSU – Stick Radio Base Station Card (Optional).....	36
3.2.6. iVIS CSU – Passive Alarm Interface Card (Optional)	36
3.2.7. iVIS CSU – Field Phone Interface Card (Optional)	37
3.3. iVIS User Unit (UU) PCBA Cards:.....	38
3.3.1. iVIS UU – iVIS 2 Knob Audio Card.....	38
3.3.2. iVIS UU – iVIS Combo Card (Optional)	38
3.4. iVIS QWERTY Keyboard Card:	40

List of Figures

Figure 1 - iVIS CSU - Isometric Top View	19
Figure 2 - iVIS CSU - Isometric Bottom View	19
Figure 3 - iVIS CSU - Ports and Peripherals Bottom View	21
Figure 4 - iVIS CSU - Ports and Peripherals Top View	22
Figure 5 - iVIS UU - Front View	24
Figure 6 - iVIS UU - Functional Keypad Keys	25
Figure 7 - iVIS UU – Ports and Peripherals	26
Figure 8 - iVIS Headset - ANR Headset Front View	28
Figure 9 - iVIS Headset - PNR Headset Front View	29
Figure 10 - iVIS - QWERTY Keyboard Unit Front View	30
Figure 11 - Stick Radio	30
Figure 12 - Stick Radio Dock Station	31
Figure 13 - iVIS Implementation for CMTV	32
Figure 14 - iVIS Implementation for AATV	33

List of Tables

Table 1 - iVIS CSU - Dimensions and Weight	21
Table 2 - iVIS CSU - Port and Peripherals	23
Table 3 - iVIS UU - Dimensions and Weight	26
Table 4 - iVIS UU - Port and Peripherals	27

List of Abbreviation

24 VDC	+24 Voltage Direct Current
AATV	Armored Ambulance Tracked Vehicle
ALNS	Automatic Land Navigation System
ANC	Active Noise Cancellation
ANR	Active Noise Reduction
ANT	Antenna
BITE	Built-In Test Equipment
BMP-II	Boyevaya Mashina Pekhoty-II
BMS	Battlefield Management System
CMDR	Commander
CMTV	Carrier Mortar Tracked Vehicle
COR	Carrier Operated Relay
CSU	Central Switch Unit
DC	Direct Current
DCH	Digital Control Harness
DRV	Driver
ETH	Ethernet
EVWS	Electronic Voice Warning System
FT	Field Telephone
GPS	Global Positioning System
GUNR	Gunner
HF	High Frequency
HS	Headset
IC	Intercom
ICV	Infantry Combat Vehicle
IP	Internet Protocol
iVIS	Intelligent Vehicle Intercom System
KBRD	Keyboard
LOS	Line of Sight
MGMT	Management
MTBF	Mean Time Between Failures
MTTR	Mean-Time-To-Repair
OLED	Organic Light-Emitting Diode
PC	Personal Computer
PCBA	Printed Circuit Board Assembly
PCM	Pulse Code Modulation
PNR	Passive Noise Reduction
PSU	Power Supply Unit
PTT	Press to Talk
PWR	Power
RAD	Radio
RIU	Radio Interface Unit
RS232	Recommended Standard 232
SMA	SubMiniature Version A

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SPE	Single Pair Ethernet
TKS	Tank System
UHF	Ultra-High Frequency
USB	Universal Serial Bus
UU	User Unit
VAD	Voice Activity Detection
VDC	Voltage Direct Current
VHF	Very High Frequency
VOL	Volume

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Chapter 1 – Brief Technical Specifications

This chapter provides an overview of the technical characteristics of the iVIS system. It describes the major hardware and functional sub-systems, outlining their roles, capabilities, and interface requirements. Detailed technical specifications for each sub-system are presented in the subsequent sub-sections.

1.1. About iVIS

The iVIS (Intelligent Vehicle Intercom System) is designed to support multiple voice communication scenarios among several interconnected units. The system primarily operates in pre-configured modes while also supporting dynamic configuration changes during runtime. This document describes the principle of operation, and the hardware and software modules required to meet the system requirements.

The iVIS system provides a robust and reliable communication solution for use within combat vehicles. It is designed to interface with onboard communication equipment such as radios, as well as vehicle sensors and auxiliary systems. The solution is ruggedized for vehicular environments and operates using vehicle-supplied power.

All iVIS system components are interconnected via an IP-based network operating within a common network domain. Communication between system components is carried out over IP for the exchange of audio streams, control signals, status information, and system events.

1.2. Core Functionalities and Features of iVIS

1. iVIS supports both voice and data communication, allowing for flexible mission applications and future system enhancements.
2. The system can handle, route, and switch a minimum of four simultaneous voice and data channels, ensuring smooth communication across all crew stations and connected systems.
3. It enables simultaneous transmission and operation of at least four HF/VHF/UHF radio sets based on the latest digital communication technologies.
4. It ensures zero interference between the intercom and radio networks, even when used concurrently.
5. The system is programmable in the field, allowing configuration updates and software upgrades without hardware replacement.
6. Seamless integration with a wide range of systems, including tactical and commercial radios (HF/VHF/UHF), navigation systems (GPS/ALNS), electronic voice warning systems, and other in-vehicle electronics, accessible via the crew's control harness.
7. Crew units include alphanumeric displays for short messages or command prompts, enhancing coordination and situational awareness.
8. Voice Channels are digitized at 128kbps PCM.
9. Data rate of data channels can be configured between 1200 bits/s to 115.2 kbps.

Advanced noise reduction algorithms and auto push-to-talk (PTT) functionality enable clear, hands-free voice communication, even in high-noise environments.

1.3. Components of iVIS System

This section provides a detailed description of all hardware components that constitute the iVIS system. Each component is presented under its own sub-heading and includes information on physical construction, external interfaces, operational characteristics, and mechanical dimensions.

The components listed in this section collectively form the complete iVIS architecture and support its communication, power distribution, processing, and system management functions.

More than ten individual units are described, including electronic modules, user interface units, cabling assemblies, power modules, and antenna-related components. The following sub-sections outline the technical specifications, dimensional details, connector definitions, and functional roles of each component for ease of integration, installation, and maintenance.

Various components of iVIS system are given below as:

1. Central Switch Unit (CSU)
2. User Unit (UU)
3. ANR Headset
4. Cable/Harness Set
5. QWERTY Keyboard Unit
6. Stick Radio (Handheld)

1.4. Design & Integration Characteristics

1. iVIS is designed to occupy minimal space within the fighting compartment, ensuring no obstruction to crew movement or ergonomics.
2. The system is easily integrated into both existing and future armored vehicle platforms, including tanks, ICVs, and their variants.
3. Installation is field executable within 6 hours, without the need of removal of turrets or major vehicle assemblies, ensuring minimal disruption to operational readiness.
4. The system is modular and configurable, allowing adaptation to specific vehicle roles and mission profiles without altering core architecture.
5. In-situ repair capability allows for quick replacement of faulty modules, supporting efficient maintenance and maximum system uptime.
6. The system enables full intercommunication within the crew of TKS and BMP-II, including stick members of BMP-II, facilitating seamless communication between the commander, gunner, driver, section commander, section 2IC and dismounted troops (sticks).
7. Overall system peak power consumption of the inter-crew communication components is approx. 48 Watts, with a nominal working power consumption of around 30 Watts.
8. Mean Time Between Failures (MTBF) is more than 12,000 hours. Mean-Time-To-Repair (MTTR) is less than 6 hours.
9. The iVIS components are sealed against dust, humidity, fungi and insects during storage and usage conditions. The exposed parts are resistant to corrosive actions of the atmosphere.
10. It is fully capable of interfacing with Battlefield Management System (BMS), making it future-ready for integrated battlefield applications.

1.5. iVIS Central Switch Unit (CSU)

The iVIS Central Switch Unit (CSU) is a multi-card system comprising several functional cards, each responsible for specific functions such as power conditioning, networking, processing, and radio interfacing within the CSU. Together, these cards operate cohesively to deliver the complete power distribution, communication, and radio integration capabilities required by the iVIS architecture.

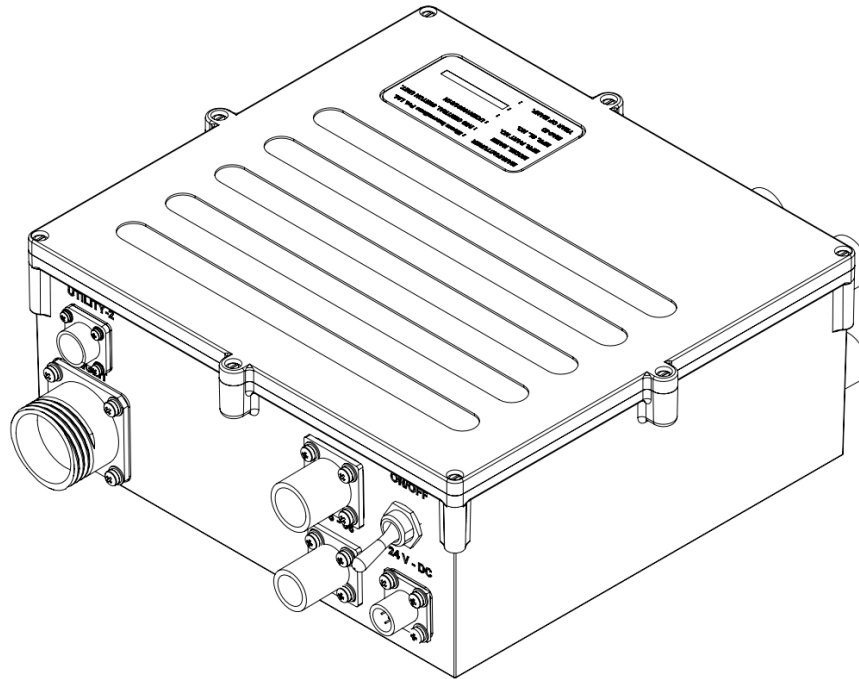


Figure 1 - iVIS CSU - Isometric Top View

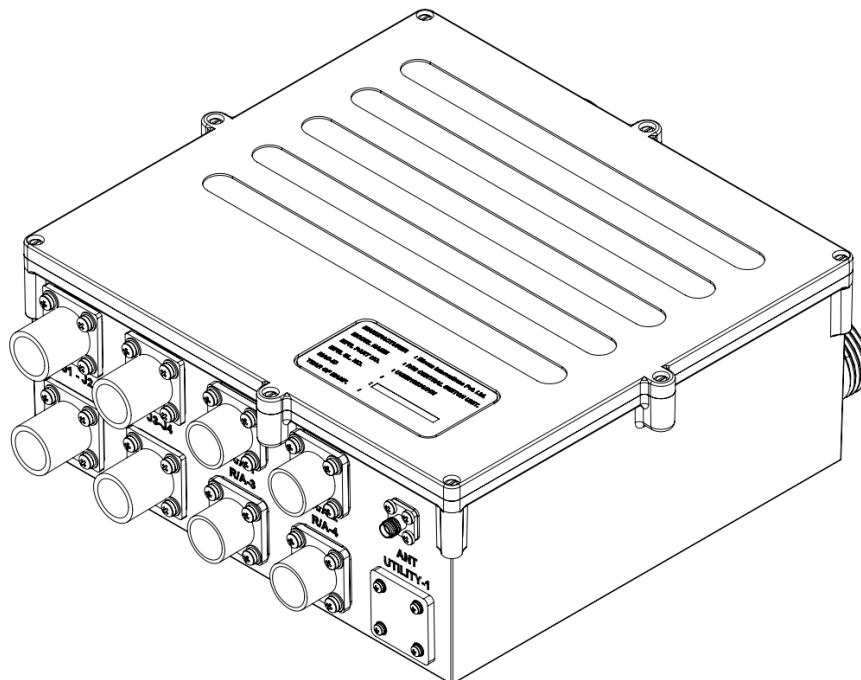


Figure 2 - iVIS CSU - Isometric Bottom View

1.5.1. iVIS CSU – Key Functions and Capabilities:

1. Centralized Power Distribution

The Central Switch Unit (CSU) receives power directly from the vehicle power source and distributes power to all connected iVIS Units through the Ethernet & Power ports and the Utility port.

By functioning as a centralized power bridge, the CSU ensures stable, efficient, and reliable power delivery across the entire iVIS architecture, supporting continuous operation of all interconnected sub-systems.

2. System Networking and Communication Bridge

The Central Switch Unit (CSU) incorporates multiple Ethernet interfaces and functions as the primary network aggregation and switching unit for the iVIS system.

All iVIS User Units interconnect through the CSU via five combined Standard Ethernet and Power ports with one Combined Single Pair Ethernet and Power Port, enabling reliable data communication, control signaling, and system-wide synchronization across all connected sub-systems. This centralized networking architecture ensures efficient traffic routing, reduced cabling complexity, and consistent communication performance throughout the iVIS system.

3. Radio Interface Support (Up to 4 Radios)

The Central Switch Unit (CSU) supports connectivity for up to four (4) in-vehicle radio sets, enabling simultaneous integration and operation of multiple radios within the iVIS system.

Each radio interface provided by the CSU includes:

- a. Dedicated Audio/AUX interface for transmitting and receiving audio paths for all radios.
- b. Optional Data interface for radio messaging, control, and status exchange for all radios.

The Audio/AUX interfaces facilitate half duplex audio communication between the in-vehicle radios and the iVIS system, supporting voice transmission and reception. The data interfaces are used for exchanging control commands, messaging data, and operational status information over the radios and the iVIS subsystems.

All connected radios can be accessed and utilized by any authorized User Unit present within the iVIS system, allowing flexible radio selection, shared radio resources, and seamless communication across multiple users and operational roles.

4. Built-in Base Station

The Central Switch Unit (CSU) incorporates an integrated stick radio base station. The CSU is equipped with an SMA connector for an external VHF antenna, enabling connection to a suitable antenna for dismounted soldier communication with an effective range of up to approximately 1500 meters in LOS mode.

5. Management and Maintenance Interface

For configuration, diagnostics, and maintenance, the CSU provides a dedicated RJ45 Management (MGMT) Ethernet connector. A laptop or PC can be connected to this interface to access system management tools, perform configuration updates, monitor system health, and carry out troubleshooting tasks.

6. Utility Interfaces

The iVIS system provides multiple optional utility interfaces through the Utility Port on the Central Switch Unit (CSU) for integration with external devices and vehicle systems. If equipped, the CSU supports data interface protocols for connecting peripherals such as the Docking station and External Alarm Unit, and for interfacing with other vehicle-mounted electronic sub-systems.

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If equipped, the Utility Port also offers an audio output for an external loudspeaker and power output for any of the utility devices from the CSU power distribution.

In addition, if equipped, the CSU provides dedicated digital input interfaces for connecting external digital alarm sources and supports Magneto Field Telephone connectivity for integrating legacy analog communication devices.

1.5.2. iVIS CSU – Dimensions and Weight:

Sr. No.	Dimension and Weight Description	Measure (Tolerance)
1	Length	210 mm (± 1 mm)
2	Height	197 mm (± 1 mm)
3	Depth	86 mm (± 1 mm)
4	Weight	2.250 Kg (+/- 200 grams)

Table 1 - iVIS CSU - Dimensions and Weight

For the ease of understanding, dimensions are taken considering unit is wall mounted & all connectors are Up or down facing.

1.5.3. iVIS CSU – Ports and Peripherals

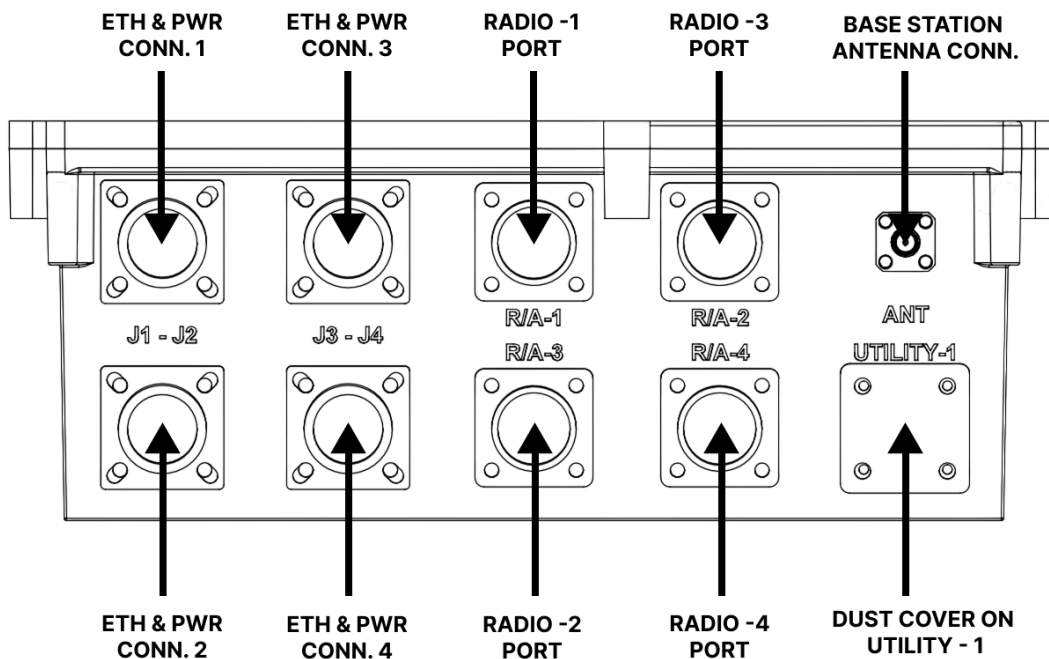


Figure 3 - iVIS CSU - Ports and Peripherals Bottom View

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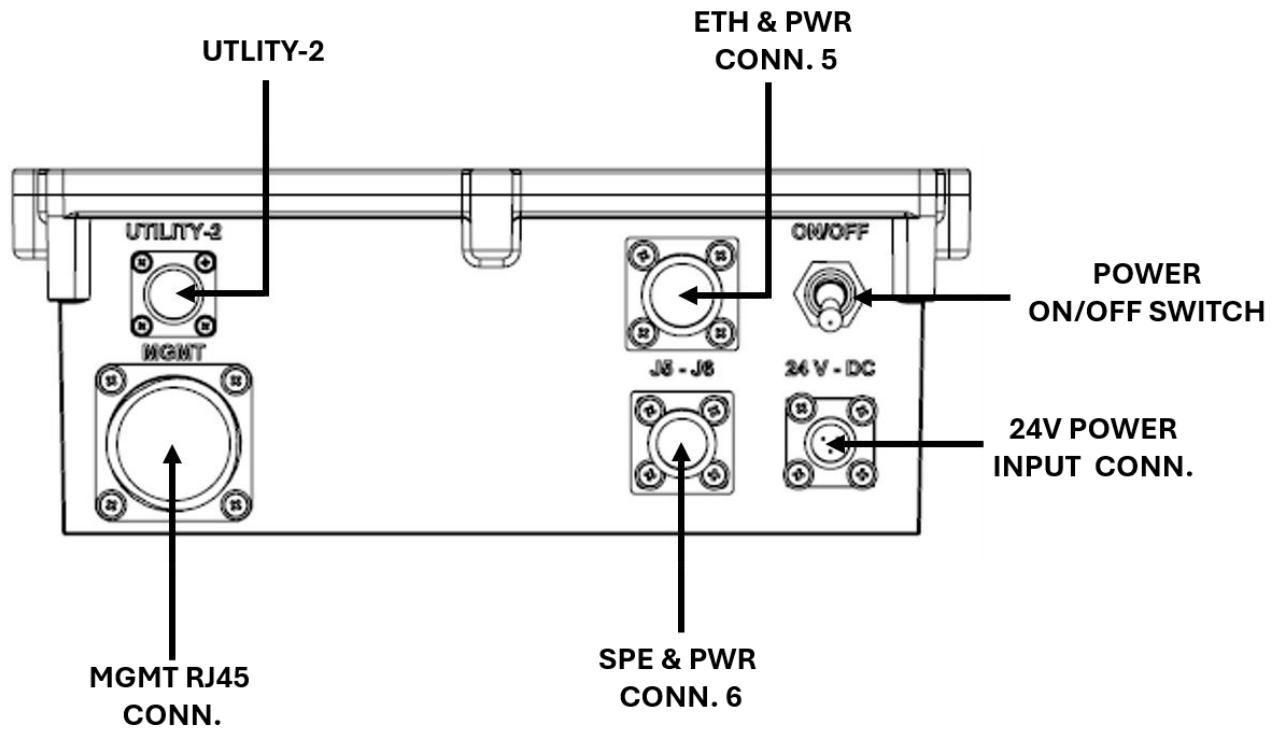


Figure 4 - iVIS CSU - Ports and Peripherals Top View

Sr. No.	Port Name	Qty	Description
01	R/A-1	04	Radio Ports
02	R/A-2		
03	R/A-3		
04	R/A-4		
05	J1	05	Ethernet and Power Connector
06	J2		
07	J3		
08	J4		
09	J5		
10	J6	01	SPE(Single Pair Ethernet) & Power Connector
11	MGMT	01	Management and Maintenance Port
12	ANT*	01	Stick Radio Base Station Antenna Port
13	UTILITY-1*	02	Utility Ports
14	UTILITY-2*		
15	24 V-DC	01	Power Input Connector
16	On/Off	01	System Power On/Off Switch

Table 2 - iVIS CSU - Port and Peripherals

***NOTE1:** ANT, UTILITY-1, and UTILITY-2 ports are optional. These ports are populated based on project specific requirement if not populated they will be covered with dust covers.

***NOTE2:** ANT and UTILITY-2 ports are populated for the DCH Project.

1.5.4. iVIS CSU – Interfaces:

1. Power Interface

Power Input – 1 No. (24 V DC)

Primary 24 V DC input (range 12–36 V DC) from the vehicle power supply.

Power Switch – 1 No. (ON/OFF)

Controls power to the entire iVIS system.

2. Radio Interfaces

Radio Ports – 4 Nos. (R/A-1, R/A-2, R/A-3, R/A-4)

Used for connection to qty 4 numbers of in-vehicle radio sets, supporting Audio/AUX and other services.

3. ETH and Power Interfaces

ETH & Power Ports – 5 Nos. (J1, J2, J3, J4, J5)

Used to connect User Units (UUs), providing combined Ethernet data and power.

SPE & Power Port – 1 No. (J6)

Provides Single Pair Ethernet (SPE) based ETH data and power connectivity to a User Unit using reduced wiring.

4. Management Interfaces

Management & Maintenance Port – 1 No. (MGMT)

1. RJ45 Ethernet interface for system configuration, diagnostics, and maintenance via laptop or PC.
2. MGMT port can be used for interfacing with other vehicle equipment (e.g. BMS, EVWS) over slandered ethernet.

5. Utility Interfaces - Optional

Utility Interfaces 1 - 1 No. (Utility-1)

Supports maximum 8 digital alarm inputs and send alerts on User Units. Through this, if equipped one number of Magneto Field Telephone can be connected.

Utility Interface 2 – 1 No. (Utility-2)

Multi-function interface providing various data interfaces, Speaker output, and Power, used for external subsystem integration like Loudspeaker Unit and with other vehicle equipment (e.g. ALNS/GPS) over RS232.

6. Stick Radio Base Station Interfaces – Optional

Stick Radio Base Station Antenna Port – 1 No. (ANT)

SMA connector for connecting an external antenna for the embedded stick radio base station.

1.6. iVIS User Unit (UU)

The iVIS User Unit (UU) serves as the primary crew interface within the iVIS system. It enables headset-based communication, user interaction, and real-time access to system and operational information. The UU is designed to provide intuitive control, enhanced situational awareness, and reliable communication for personnel operating in in-vehicle or fixed installations.

Depending on system configuration and role assignment, the User Unit can function as a Commander Unit, Driver Unit, Gunner Unit, or general Crew Unit. Each UU can be configured to provide role-specific audio routing, control permissions, and access to system functions, ensuring that every crew member receives the appropriate level of control and information required for their operational role.

By supporting multiple crew roles through a common hardware platform, the iVIS User Unit enables flexible system deployment, simplified logistics, and consistent user experience across all crew positions.

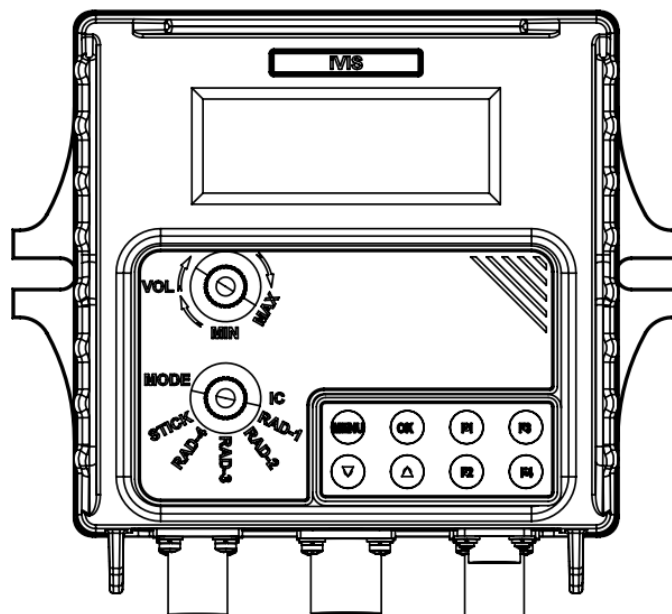


Figure 5 - iVIS UU - Front View

1.6.1. iVIS UU – Key Functions and Capabilities

1. Primary Operator Interface

The User Unit functions as the main human machine interface of the iVIS system, allowing operators to access communication channels, system status, and operational features in real time. It enables direct interaction with radios, Stick Radio services, and other iVIS subsystems through CSU.

2. Headset-Based Communication and Control

The User Unit supports headset-based communication and includes dedicated front-panel control knobs for:

1. Headset speaker volume control, allowing precise adjustment of audio output levels
2. Mode selection, enabling the user to select the desired runtime communication interface or operational mode

These controls provide quick, direct single step and intuitive access to critical communication functions during operation. These controls support blind operation modes by implementing tones and voice prompts on knob position change.

3. User Selectable Communication Modes

The User Unit allows the operator to select different communication modes using the dedicated mode selection knob. These modes define the active audio and communication path for the headset, enabling flexible interaction with in-vehicle and dismounted communication systems. The available modes are:

- (a) **IC (Intercom)**
Enables internal vehicle crew communication, allowing voice interaction between all connected User Units within the vehicle.
- (b) **RAD-1 (Radio 1)**
Select the in-vehicle radio connected to Radio Interface 1 (R/A-1).
- (c) **RAD-2 (Radio 2)**
Select the in-vehicle radio connected to Radio Interface 2 (R/A-2).
- (d) **RAD-3 (Radio 3)**
Select the in-vehicle radio connected to Radio Interface 3 (R/A-3).
- (e) **RAD-4 (Radio 4)**
Select the in-vehicle radio connected to Radio Interface 4 (R/A-4).
- (f) **STICK (Stick Radio)**
Selects the built-in Stick Radio integrated within the CSU, enabling communication with dismounted soldiers operating outside the vehicle.

These selectable modes provide seamless switching between internal crew communication, multiple in-vehicle radios, and the embedded base station, ensuring operational flexibility across different mission scenarios.

4. Display and User Feedback

The User Unit is equipped with a 4×20 alphanumeric OLED display, used to present runtime and system information such as:

- 1. Communication and channel status
- 2. Selected radio or interface
- 3. Alerts, alarms, and notifications
- 4. Operational and system messages
- 5. Run time BITE messages

This display ensures continuous situational awareness for the operator.

5. Functional Keypad for Operational Functions

The User Unit features an integrated 8-button custom keypad designed to provide fast and intuitive user interaction.

The keypad includes the following keys:

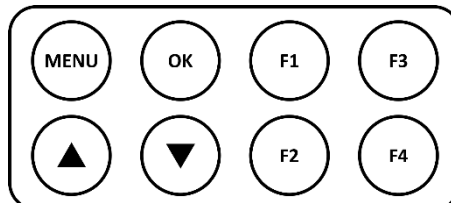


Figure 6 - iVIS UU - Functional Keypad Keys

The MENU, OK, and arrow keys are used for navigating system menus, viewing system information, and accessing configuration and diagnostic screens, including Built-In Test Equipment (BITE) diagnostics.

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The function keys (F1 to F4) provide configurable shortcut access to frequently used operational functions such as:

1. Quick access to location display (If connected to ALNS system)
2. Alarm acknowledgment and alert handling (if interfaced with alarms)
3. User-defined system shortcut functions

By separating quick-access operational controls from menu-based diagnostic functions, the keypad design ensures both operational efficiency and safe access to maintenance and diagnostic features, reducing the risk of accidental activation during mission-critical operations.

6. Centralized Power and Network Connectivity

The User Unit receives both power and Ethernet connectivity directly from the CSU through the Ethernet & Power port or the SPE & Power port. This centralized architecture simplifies installation, reduces cabling, and ensures reliable communication and power delivery across the iVIS system.

1.6.2. iVIS UU – Dimensions and Weight:

Sr. No.	Dimension and Weight Description	Measure (Tolerance)
1	Length	147 ± 1 mm
2	Length with mounting cuts	180 ± 1 mm
3	Height without protective hinges	142 ± 1 mm
4	Height with protective hinges	157 ± 1 mm
5	Depth	84 ± 1 mm
6	Weight	1.5 Kg ± 100 gm

Table 3 - iVIS UU - Dimensions and Weight

For ease of understanding, dimensions are taken considering unit is wall mounted & all connectors are down facing.

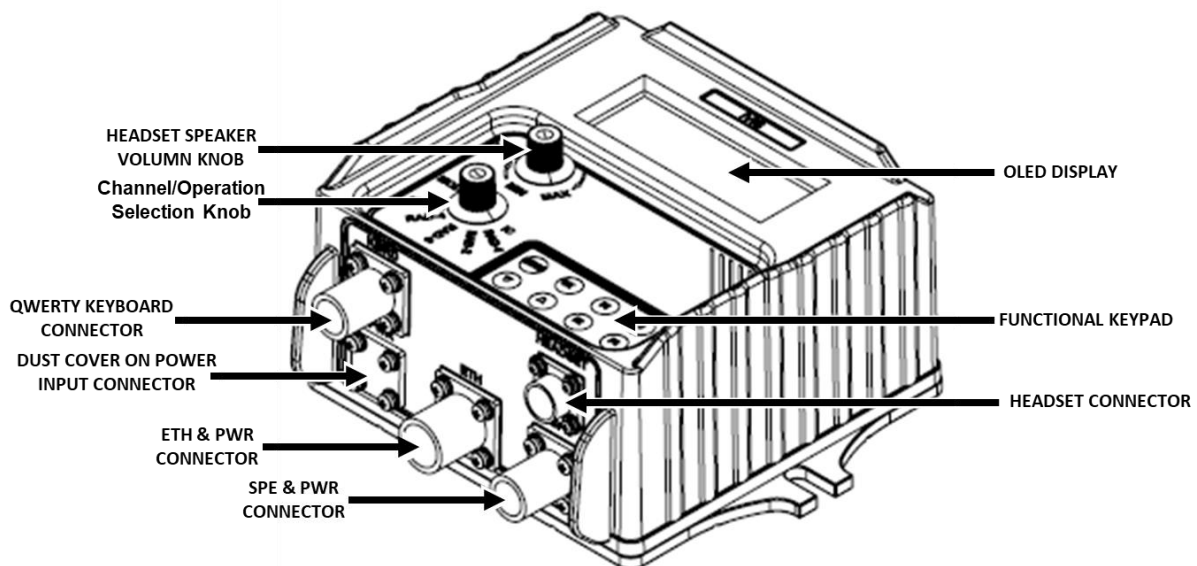


Figure 7 - iVIS UU – Ports and Peripherals

Sr. No.	Port Name	Qty	Description
1	ETH	01	Ethernet & Power Connectors
2	SPE*	01	SPE(Single Pair Ethernet) & Power Connector
3	HEADSET	01	ANR/PNR Headset Port
4	KBRD*	01	QWERTY Keyboard Port
5	PWR*	01	Power Input Port
6	-	01	Display
7	VOL	01	Headset Speaker Volume Knob
8	MODE	01	Channel Operation Knob
9	-	01	Functional Keypad

Table 4 - iVIS UU - Port and Peripherals

***NOTE:** SPE, KBRD, and PWR ports are optional. These ports are generally covered with dust covers, and the connectors are populated only based on project-specific requirements.

1.6.3. iVIS UU – Interfaces:

1. Ethernet and Power Interface

Ethernet & Power Port – 1 No. (ETH)

Provides combined Ethernet data and power connectivity from the CSU/Other UU to the User Unit using a single cable.

Single Pair Ethernet & Power Port – 1 No. (SPE)

Provides an alternative SPE-based data and power connection to the CSU/other UU when the standard Ethernet & Power port is not used, enabling reduced cabling with equivalent functionality.

2. Audio Interface

Headset Port – 1 No. (HEADSET)

Interface for connecting an ANR headset with PTT, supporting user audio input (microphone) and output (speaker).

3. User Input Interfaces (Optional)

Keyboard Port – 1 No. (KBRD)

Provides connectivity to the iVIS QWERTY Keyboard Unit for enhanced text input, messaging, and advanced user interaction.

4. Power Interface (Optional)

Vehicle Power Input – 1 No. (PWR)

Optional vehicle power input allowing the User Unit to be powered directly from the vehicle electrical system. This input provides power redundancy and supports hull–turret deployment scenarios, where separate power sources are available for hull and turret-mounted equipment.

1.7. iVIS Headset Options

1.7.1. iVIS Headset – ANR Headset

The iVIS User Units support Active Noise Reduction (ANR) / Active Noise Cancellation (ANC) headsets to ensure clear and reliable voice communication in high-noise military vehicle environments. The ANR headset actively reduces ambient and vehicle noise, improving speech intelligibility and operator comfort during prolonged operation.



Figure 8 - iVIS Headset - ANR Headset Front View

1. PTT Switch Functionality

The headset Push-To-Talk (PTT) switch supports the following operating modes:

- **Latching:** Activates voice in Intercom Mode for continuous communication.
- **OFF: Disables** audio transmission (Receive only mode).
- **Momentary:** Enables audio transmission based on the channel selected using the channel selection knob (Intercom/Radio/Stick etc).

2. Noise Control and Listening Modes

The ANR headset supports dual listening modes to balance noise reduction and situational awareness that can be switched using mode toggle switch located on the right earcup of headset.

Headset modes are as follows:

- **Forward Latch:** Passthrough Mode. Allows controlled ambient sound pickup while maintaining communication capability.
- **Centre position:** Off (Behaves like PNR Headset)
- **Backward Latch:** ANR Mode. Actively suppresses ambient and vehicle noise to enhance speech clarity.

1.7.2. iVIS Headset – PNR Headset

The iVIS User Units also support Passive Noise Reduction (PNR) headsets, providing reliable communication through mechanical noise attenuation without the use of active electronics. PNR headsets are suitable for applications where passive noise isolation is sufficient or where simplified headset operation is required.



Figure 9 - iVIS Headset - PNR Headset Front View

1. PTT Switch Functionality

The Push-To-Talk (PTT) switch functionality for PNR headsets is identical to that of ANR headsets:

- **Latching:** Activates voice in Intercom Mode for continuous communication.
- **OFF: Disables** audio transmission (Receive only mode).
- **Momentary:** Enables audio transmission based on the channel selected using the channel selection knob (Intercom/Radio/Stick etc).

2. Audio Configuration

PNR headsets support monaural and binaural audio operation, allowing flexible deployment across different crew roles and vehicle configurations.

3. Power Characteristics

PNR headsets do not require active power for noise reduction and operate without ANR or Talk-Through functionality. Audio transmission and reception are supported directly through the iVIS User Unit interface.

4. Compatibility

The iVIS system supports third-party Passive Noise Reduction headsets, enabling integration with standard military and mission-specific audio equipment.

1.8. QWERTY Keyboard

The iVIS QWERTY Keyboard Unit is an optional user input device that enhances text entry and command input capabilities within the iVIS system. It connects directly to the User Unit (UU) through the Keyboard (KBRD) Port, enabling seamless, plug-and-play integration without additional configuration.

The keyboard allows users to efficiently compose and edit text messages, significantly improving usability for messaging, data entry, and command execution when compared to keypad-based input.

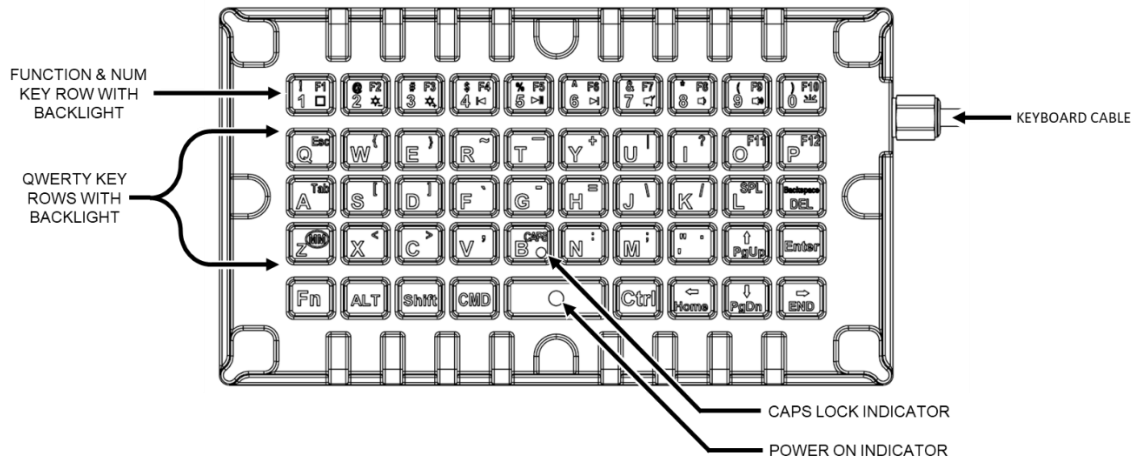


Figure 10 - iVIS - QWERTY Keyboard Unit Front View

In addition to full alphanumeric input, the iVIS QWERTY Keyboard Unit supports all functions available on the User Unit's functional keypad, ensuring complete operational control through a single device. It also provides extended functionality through dedicated function keys (F1 to F10), which can be assigned to advanced operations, shortcuts, or system-specific commands as defined by the iVIS configuration.

By combining full-text input with extended function key support, the iVIS QWERTY Keyboard improves operator efficiency, reduces input time, and enhances overall interaction with the iVIS system during both routine and mission-critical operations.

1.9. Stick Radio

iVIS supports a dedicated UHF Stick Radio to enable seamless communication between the vehicle crew and dismounted personnel operating outside the vehicle. Stick Radio communicates through the vehicle-mounted antenna interface, providing reliable line-of-sight voice communication over distances exceeding 1500 meters, depending on terrain and operating conditions.

The Stick Radio is a compact handheld transceiver featuring programmable channels, Push-to-Talk (PTT) operation, adjustable volume control, channel selection, and status indication. It operates on rechargeable battery power and is stored in a dedicated Stick Radio Dock Station inside the vehicle, where it is securely held and charged whenever not in use. This arrangement ensures that the radio remains fully charged and readily available for rapid deployment during operational missions.

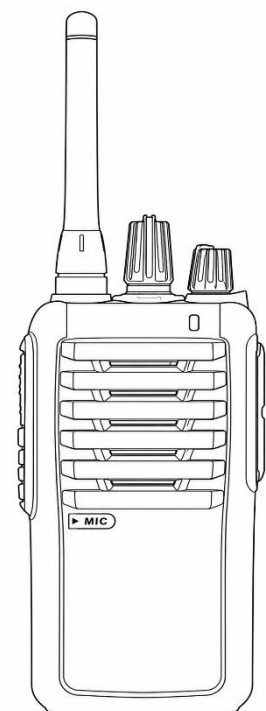


Figure 11 - Stick Radio

1.10. Stick Radio Dock Station

The Stick Radio Dock Station serves as the charging and storage unit for Stick Radio. It receives power directly from the vehicle power supply and provides the required power for battery charging. The dock securely holds the radio during vehicle operation while ensuring the battery remains charged and ready for immediate use.

The Stick Radio Dock Station enables convenient charging without removing the battery from the radio and provides a dedicated location for safe storage when the Stick Radio is not in use. It is designed for reliable operation under vehicle operating conditions and ensures the radio is readily available for deployment.

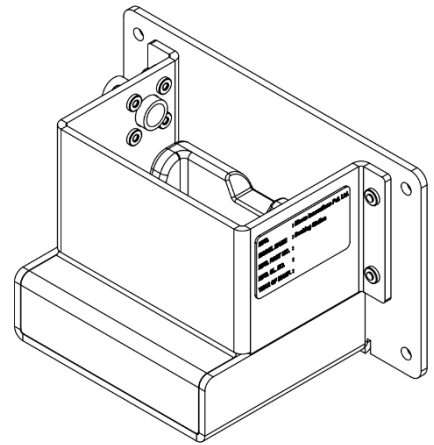


Figure 12 - Stick Radio
Dock Station

Chapter II – Brief Principles of Operation

The iVIS (Intelligent Vehicle Intercom System) operates as an integrated, IP-based intercom and radio interface system for military vehicles. The system architecture, as shown in the Functional Block Diagram, consists of multiple User Units, a Central Switch Unit (CSU), radio interfaces, headsets and associated power and antenna interfaces.

The Central Switch Unit (CSU) functions as the core node of the system. It distributes vehicle power, manages Ethernet-based communication, and performs IP packet routing between connected User Units, radio equipment, and external audio devices. The CSU interfaces with multiple radios (RAD-1, RAD-2, RAD-3) through dedicated radio interface ports, enabling inter-crew and radio communication.

Each User Unit (DVR Unit, GNR Unit, SEC-CDR Unit, CDR Unit, SEC-2IC Unit, etc.) is connected to the CSU via combined ethernet power connectors. User headsets are directly connected to the User Units, allowing crew members to communicate internally or transmit audio to selected radios using the Push-To-Talk (PTT) controls. The operating mode (intercom or radio) is determined by the user's channel selection and PTT operation.

Audio signals captured from the headsets are digitized at the User Units and transmitted over the internal Ethernet network to the CSU. Switching of voice occurs in the network avoiding dependency on any single common voice switch. Voice streams switching is handled intelligently by each USER Unit/Radio Port based on the selected configuration and operational mode, of self and other User Units. Each User Unit/Radio port selects the IP voice stream from the network and routes the audio to Headset of User Units (intercom communication) or to one of the connected radios, or to external audio devices such as the Speaker Unit.

The system also supports auxiliary equipment including a Keyboard Unit for control and configuration, a Stick Radio with a dedicated VHF antenna, and a Charging Station, all powered from the vehicle's 24 V DC supply. Antenna connections are provided for radio communication, while standard Ethernet interfaces are used for internal data exchange.

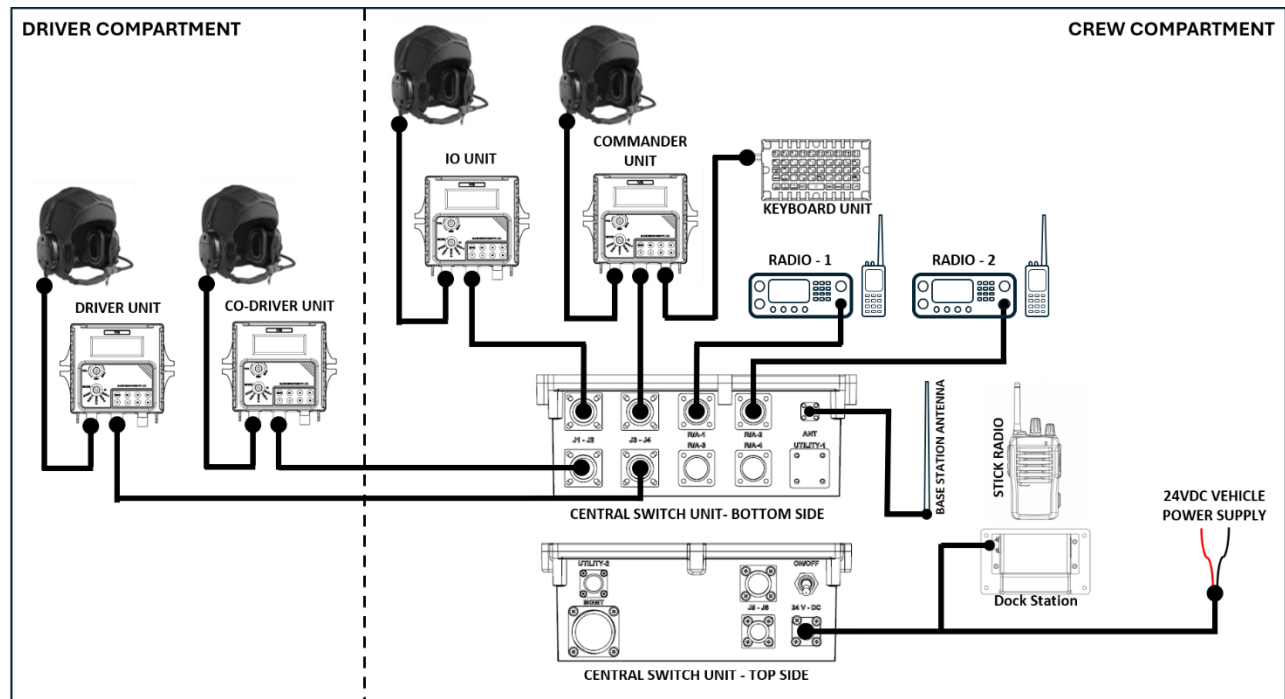


Figure 13 - iVIS Implementation for CMTV

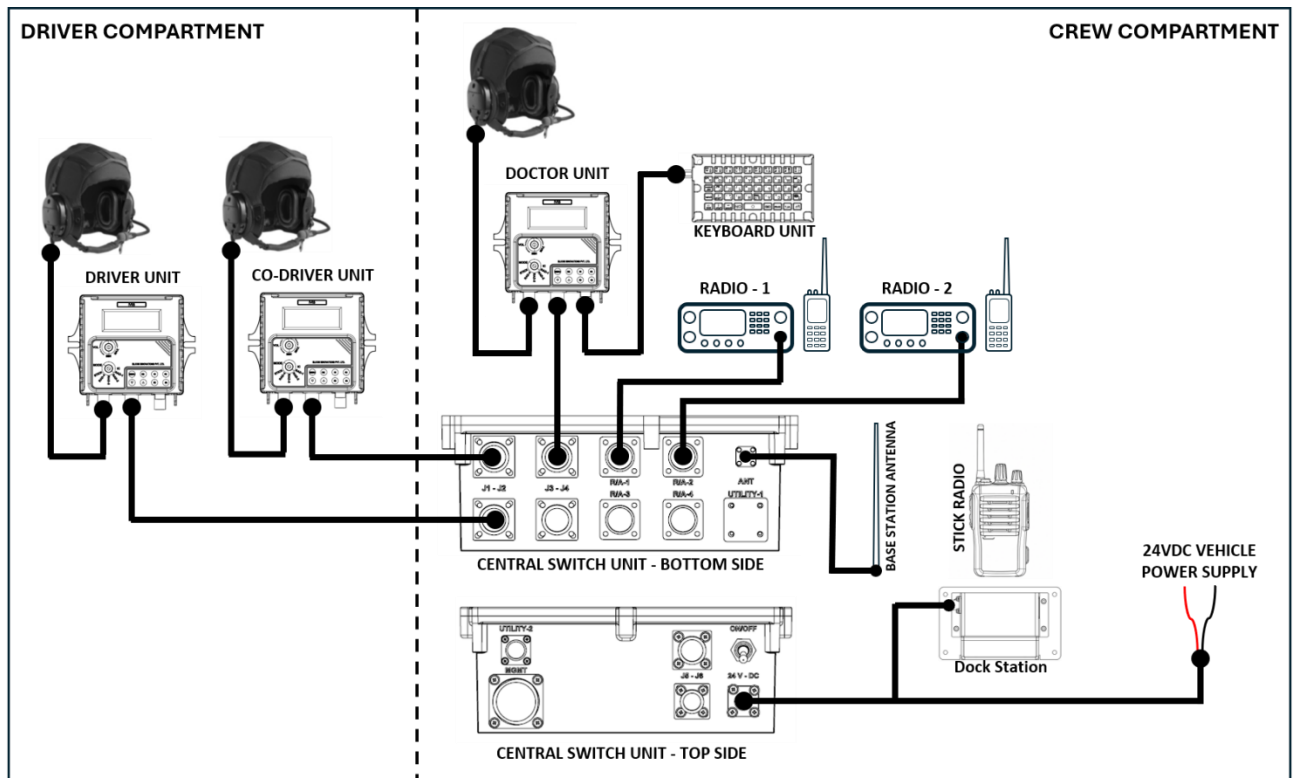


Figure 14 - iVIS Implementation for AATV

All iVIS components are powered from the vehicle electrical system and interconnected through a common IP network. The system is designed to operate in pre-configured modes while allowing runtime configuration changes without disrupting ongoing communication. Protective features and filtering ensure reliable operation in harsh vehicular and electromagnetic environments.

Chapter III – Detailed Circuit Theory

This chapter provides a detailed explanation of the circuit-level design of the iVIS system. It covers the theoretical operation of individual circuits, component selection rationale, and signal processing mechanisms. Detailed circuit descriptions and analyses are organized in the subsequent sub-sections.

3.1. About iVIS PCBA Cards:

The iVIS PCBA cards form the core hardware building blocks of the iVIS system. Each PCBA is designed to perform specific functional roles such as audio processing, control, networking, power management, and interface handling. Together, these cards enable reliable voice communication and system control within the vehicle environment.

The PCBA cards are designed to operate in a distributed ethernet-based IP network architecture, allowing seamless communication between modules over the internal network. They interface with external peripherals such as headsets, radios, vehicle sensors, and control units as required by the system configuration.

All iVIS PCBA cards are engineered for rugged operation, vehicle power compatibility, and compliance with environmental and operational requirements typical of combat vehicle applications. Detailed descriptions of individual PCBA cards, their functions, and interfaces are provided in the subsequent sections.

3.2. iVIS Central Switch Unit (CSU) PCBA Cards:

The iVIS Central Switch Unit (CSU) is a multi-card system comprising several functional cards, each responsible for specific functions such as power conditioning, networking, processing, and radio interfacing within the CSU. Together, these cards operate cohesively to deliver the complete power distribution, communication, and radio integration capabilities required by the iVIS architecture.

The CSU consists of the following cards:

- a. iVIS EMI-EMC Power Filter Card
- b. iVIS 10 Port Ethernet Switch with SPE Card
- c. iVIS Dual Radio Card
- d. iVIS Dual Radio Extension Card (Optional)
- e. Stick Radio Base Station Card (Optional)
- f. Passive Alarm Interface Card (Optional)
- g. Field Phone Interface Card (Optional)

3.2.1. iVIS CSU – iVIS EMI/EMC Power Filter Card

Provides input power conditioning and EMI/EMC filtering to protect the iVIS system from electrical noise and disturbances originating from the vehicle power source, ensuring stable, reliable, and compliant 24 V power delivery to all iVIS subsystems.

The card accepts a 24 V DC input (range 16–32 V DC) from the vehicle supply and distributes 24 V DC power through output channels, as detailed below:

- Outputs supply 24 V power to the Ethernet & Power Ports
- Output supply 24 V power to the Utility Port
- Output is used internally within the CSU for iVIS Dual Radio Card.

This card performs the essential functions of EMI/EMC protection and filtering, initial surge protection and centralized 24 V power distribution for the entire iVIS system.

3.2.2. iVIS CSU – 10-Port Ethernet Switch with SPE Card

This card implements the core Ethernet switching functionality of the CSU, providing both standard Ethernet and Single Pair Ethernet (SPE) connectivity for the iVIS system.

It supports maximum up to 10+1 ports: 10 standard Ethernet ports and 1 SPE port.

- a. The 10 Ethernet ports support 10/100/1000Base-T (Gigabit Ethernet) and enable high-speed data communication across the system.
- b. The SPE port supports 10Base-T1S, allowing multidrop communication over a single twisted pair, enabling simplified cabling to connected subsystems.

The uses of Ethernet ports as below:

- Five ports connect to Ethernet & Power Ports for subsystem connectivity.
- One port is used internally within the CSU to interface with the iVIS Dual Radio Card.
- One port is used for the Management RJ45 (MGMT) interface.

The SPE port is routed to the SPE Port, providing combined data and power delivery to iVIS subsystems over a minimal two-pair wiring setup.

This card serves as the central communication hub of the CSU, enabling high-speed, reliable networking and efficient integration of iVIS subsystems while minimizing cabling complexity.

3.2.3. iVIS CSU – iVIS Dual Radio Card

The iVIS Dual Radio Card provides interfaces for two in-vehicle radio sets, enabling full integration of radio audio, messaging, and control within the iVIS system.

Each radio interface includes:

- Isolated Audio/AUX connection for transmit and receive audio.
- Isolated PTT/COR signaling.
- Isolated data interface for messaging, control commands, and status exchange.

The card provides isolated Audio/AUX and data interfaces for the following Radio Ports:

- Radio Interface 1 (R/A-1)
- Radio Interface 2 (R/A-2)

The card features a USB interface for direct connectivity with the iVIS 10-Port Ethernet Switch with SPE Card, enabling configuration, monitoring, and management of the Ethernet switching functions.

It also provides a speaker output interface routed to the CSU Utility Port for connection to an external loudspeaker unit, along with an 8-channel Alarm input interface for connection to the Alarm Interface Card, enabling centralized alarm and alert handling within the CSU.

It also provides a stacking provision for the iVIS Dual Radio Extension Card, allowing the addition of further isolated Audio/AUX and data interfaces to support additional radios and the Stick Radio Base Station Card.

This card has its own isolated Power Supply Unit (PSU), ensuring stable operation and electrical separation from other CSU modules.

iVIS Dual Radio Card has the main processing within the CSU, the iVIS Dual Radio Card can process all four radio audio channels, serial interfaces, and base station signals, enabling seamless integration, routing, and coordination of radio communications across the iVIS system.

By centralizing radio management while maintaining electrical isolation, this card ensures efficient half duplex audio communication, reliable data handling, and scalable expansion for additional radios, alarms and other connectivity through expansion card.

3.2.4. iVIS CSU – Dual Radio Extension Card (Optional)

The iVIS Dual Radio Extension Card expands the radio connectivity of the CSU by providing interfaces for two additional in-vehicle radios, enabling support for up to four radios within the iVIS system.

This card is stacked directly onto the iVIS Dual Radio Card and functions purely as an expansion module. All signal processing, control logic, and power supply are provided by the iVIS Dual Radio Card. The Dual Radio Extension Card itself does not perform independent processing.

The card provides isolated Audio/AUX and data interfaces for the following Radio Ports:

- Radio Interface 3 (R/A-3)
- Radio Interface 4 (R/A-4)

In addition, the extension interfaces also support connectivity for the Stick Radio Base Station when integrated into the CSU.

The Dual Radio Extension Card further provides various types of serial data interface provisions for system and vehicle electronics/subsystem integration, along with a dedicated interface for connecting the Magneto Field phone Card, enabling Magneto Field Telephone (AFT) functionality within the iVIS system.

It also includes an additional 8-channel Alarm input interface for connection to additional Alarm Card, extending the CSU's alarm and alert handling capability.

By extending the Dual Radio Card, this modular design enables scalable radio integration while maintaining centralized processing, isolation, and control within the CSU.

3.2.5. iVIS CSU – Stick Radio Base Station Card (Optional)

The Stick Radio Base station Card implements base station functionality within the iVIS Central Switch Unit (CSU).

This card operates as an Analog/Digital radio module and performs all Analog/Digital radio processing, enabling direct Stick radio communication when integrated into the CSU.

This card connects electrically and mechanically to the iVIS Dual Radio Extension Card, which provides the physical connectivity and interface routing required for the base station operation. The iVIS Dual Radio Card provides system-level integration, including power management, control, and communication interfacing between the Stick Radio Base station Card and the rest of the iVIS system.

The Stick Radio Base station Card is equipped with an inbuilt SMA connector interface for connection to an external antenna, supporting reliable radio communication in base station mode.

By integrating the Stick Radio Base station Card, the CSU delivers embedded base station capability, complementing in-vehicle radios and extending the overall communication flexibility and operational capability of the iVIS system.

3.2.6. iVIS CSU – Passive Alarm Interface Card (Optional)

The Alarm Interface Card provides a dedicated interface for monitoring external alarm and status signals within the iVIS system. It is designed to connect directly to the iVIS Dual Radio Card or the iVIS Dual Radio Extension Card, enabling seamless integration with existing system.

The card supports 8 digital isolated input channels, allowing reliable detection of external events while protecting the iVIS system from electrical noise, ground loops, and voltage transients.

Each digital input can be configured to generate real-time alerts and notifications on connected User Units (UUs). These alerts enable operators to receive immediate awareness of critical system or environmental conditions.

Typical applications include, but are not limited to:

- Low fuel level indication
- CO₂ or gas level alarms
- Equipment fault or status signals
- External safety or warning system inputs

By providing electrically isolated alarm inputs with direct user notification, the Alarm Interface Card enhances situational awareness, safety monitoring, and operational responsiveness within the iVIS system.

3.2.7. iVIS CSU – Field Phone Interface Card (Optional)

The (Magneto) Field Phone Interface Card enables integration of traditional magneto-based field telephones into the iVIS communication system. The card is designed to connect directly to the iVIS Dual Radio Extension Card, allowing legacy field phone equipment to interoperate with modern iVIS System infrastructure.

The interface card supports signaling and audio interfacing required for magneto field phones, enabling call initiation, ringing detection, and voice communication through the iVIS system. This allows operators to communicate with field phone users using connected User Units (UUs) without requiring standalone field phone networks.

The card provides electrical isolation and signal conditioning to ensure safe and reliable operation when interfacing with external field phone lines, protecting the iVIS system from high-voltage magneto ringing signals and line transients.

By enabling seamless connectivity between magneto field phones and the iVIS Dual Radio Extension Card, the Field Phone Interface Card extends system interoperability, supports legacy communication assets, and enhances operational flexibility in both fixed and field deployments.

3.3. iVIS User Unit (UU) PCBA Cards:

The iVIS User Unit (UU) is a modular, multi-card assembly designed to provide user interaction, audio handling, display control, and local connectivity within the iVIS architecture. The UU comprises dedicated functional cards that together deliver audio interfacing, user controls, display management, networking, and optional utility interfaces while maintaining seamless integration with the Central Switch Unit (CSU).

The User Unit consists of the following cards:

- a. iVIS 2-Knob Audio Card
- b. iVIS Combo Card

3.3.1. iVIS UU – iVIS 2 Knob Audio Card

The iVIS 2-Knob Audio Card provides the primary user interface, audio handling, and control functionality within the iVIS User Unit (UU), enabling seamless interaction between the user and the iVIS communication system.

The card includes the following user control interfaces:

- **Volume Control Knob** for headset speaker level adjustment
- **Channel / Mode Selection Knob** for selecting the desired runtime communication mode

The card provides comprehensive audio and user interaction interfaces, including:

- **Headset Audio Interface** supporting ANR/PNR headsets for microphone input, speaker output and PTT interface
- **Display Output Interface** for driving the 4×20 alphanumeric OLED display used for runtime status, alerts, and system messages
- **Keypad Interface** for connection of the 8-button functional keypad

The card also provides the following communication and connectivity interfaces:

- **Ethernet Interface** used for connection to the Ethernet & Power Port or for interfacing with the iVIS Combo Card when daisy-chaining or SPE-based connectivity is required.
- **USB Interface** for direct connectivity to the Keyboard (KBRD) Port, enabling connection of an external iVIS QWERTY keyboard .

Power handling and distribution are supported through multiple interfaces:

- **Dual DC Power Input/Output** paths supporting both standard Ethernet & Power and Single Pair Ethernet (SPE) based power delivery and daisy-chain configurations
- **Optional Reversible Direct Vehicle Power Input**, allowing the User Unit to operate directly from vehicle power as a redundant supply or in hull–turret split-power configurations

The card also includes a stacking interface for the iVIS Combo Card, enabling expansion of networking, utility, and vehicle interface capabilities within the User Unit.

By integrating user controls, audio processing, display management, power handling, and expansion capability into a single module, the iVIS 2-Knob Audio Card serves as the operational core of the User Unit, ensuring intuitive user interaction, reliable audio communication, and flexible system integration within the iVIS architecture.

3.3.2. iVIS UU – iVIS Combo Card (Optional)

The iVIS Combo Card extends the networking and utility interface capabilities of the iVIS system, enabling flexible connectivity, daisy-chaining, and vehicle interface integration across different iVIS configurations.

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The card incorporates an integrated Ethernet switching function that provides multiple downstream network connections. It supports the following Ethernet interfaces:

- **Five standard Ethernet ports**, primarily used for User Unit daisy-chain connectivity and local network expansion (typically two ports are used in standard deployments)
- **One Single Pair Ethernet (SPE) port**, used when reduced cabling is required due to space, weight, or installation constraints, while maintaining reliable data communication and power delivery

In addition to networking, the Combo Card provides multiple utility and vehicle interface provisions.

The iVIS Combo Card is designed with a flexible stacking architecture and can be integrated in multiple system configurations:

- It can be stacked directly onto the iVIS 3-Knob Audio Card within the User Unit (UU), extending networking and utility functionality while relying on the 3-Knob Audio Card for user interaction, audio processing, and primary power handling.
- It can also be stacked onto the iVIS Dual Radio Card when used in a Radio Interface Unit (RIU) configuration, enabling Ethernet daisy-chain functionality and expanded system connectivity at radio interface locations.

In all configurations, the Combo Card functions as a networking and interface expansion module, while core processing and control are handled by the host card.

By supporting multi-role deployment across User Units and Radio Interface Units, the iVIS Combo Card enhances system scalability, simplifies cabling, and enables consistent network behavior throughout the iVIS architecture.

3.4. iVIS QWERTY Keyboard Card:

The iVIS QWERTY Keyboard Unit is implemented as a single-card assembly, integrating all keyboard control and interface functions on one dedicated PCBA.

The PCBA incorporates a 5×10 key matrix, supporting a total of 49 keys, including alphanumeric, function, and control keys required for complete system interaction.

The card includes the following visual indicators:

- Green Power Indicator – indicates that the keyboard is powered and operational
- White Caps Lock Indicator – indicates the active state of the Caps Lock function

For low-light and night-time operation, the keyboard features amber backlighting, which is automatically activated upon key activity and turns off after a fixed timeout period to minimize power consumption and visual signature.

The PCBA includes a USB interface for direct connection to the User Unit via the KBRD Port, enabling reliable communication and seamless integration with the iVIS system.

By integrating key scanning, status indication, backlighting control, and USB communication into a single PCBA, the iVIS QWERTY Keyboard Unit delivers a compact, robust, and mission-ready input solution for the iVIS architecture.

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Document Revision History

Rev.	Change	Date	Author	Approved by
PA1	Created new document with component specifications, dimensions, & weights, supporting images.	Nov 18, 2025	Sonika Singh	Prafulla Sarode
PA2	Updated images and added respected port and connectors tables. Added controls & terminals of system components	Jan 16, 2026	Sonika Singh	Prafulla Sarode
PA3	Added Technical Specification, Principal of Operation and Circuit Theory	Jan 20, 2026	Shreyash K	Prafulla Sarode
PA4	Updated Qwerty keyboard, User Unit, and CSU ports & Peripherals images	Jan 20, 2026	Sonika Singh	Prafulla Sarode
PA5	Formatting & Correction	Feb 11, 2026	Shreyash K	Prafulla Sarode
PA6	Update Content	Mar 1, 2026	Shreyash K	Prafulla Sarode
PA7	Pictures Update	Apr 22, 2026	Shreyash K	Prafulla Sarode
PA8	Formatting of Document	June 16, 2026	Kriti Mathur	Prafulla Sarode
PA9	Updated iVIS Implementation	July 8, 2026	Shreyash K	Prafulla Sarode
PA10	Addition of Stick Radio and Stick radio Dock Station	July 22, 2026	Shreyash K	Prafulla Sarode